Course Information Sheet

Data Structures and Algorithms (ICT2123)

Document No. CRS-0084 · Academic Year 2023/2024 · Semester 2

Course Details

Field	Details
Course ID	8
Course Name	Data Structures and Algorithms
Course Code	ICT2123
Credit Value	3.00
Department	Information and Communication Technology (Department ID: 1)

Course Description

This course studies the organization, management, and processing of data in memory and the design of efficient algorithms. Students analyse the time and space complexity of algorithms and work with linear structures such as lists, stacks, and queues, as well as trees, hash tables, and graphs. Algorithm design strategies, including divide and conquer, greedy methods, and dynamic programming, are introduced and compared empirically. Laboratory work requires students to implement the structures and measure the performance predicted by the analysis.

Learning Outcomes

On successful completion of this course, students will be able to:

- Analyse the time and space complexity of algorithms using asymptotic notation.
- Implement linear and non-linear data structures and select an appropriate structure for a task.
- Apply design strategies such as divide and conquer, greedy methods, and dynamic programming.
- Empirically evaluate and compare algorithms against their theoretical behaviour.

Assessment and Grading

Component	Weighting	Description
Assignments and Quizzes	20%	Graded coursework issued and completed during the semester
Mid-Semester Test	10%	Written test covering the first half of the course content
Final Examination	70%	End-of-semester written examination

Note: this is a sample document generated for project testing purposes. Assessment weights and descriptive text are illustrative.